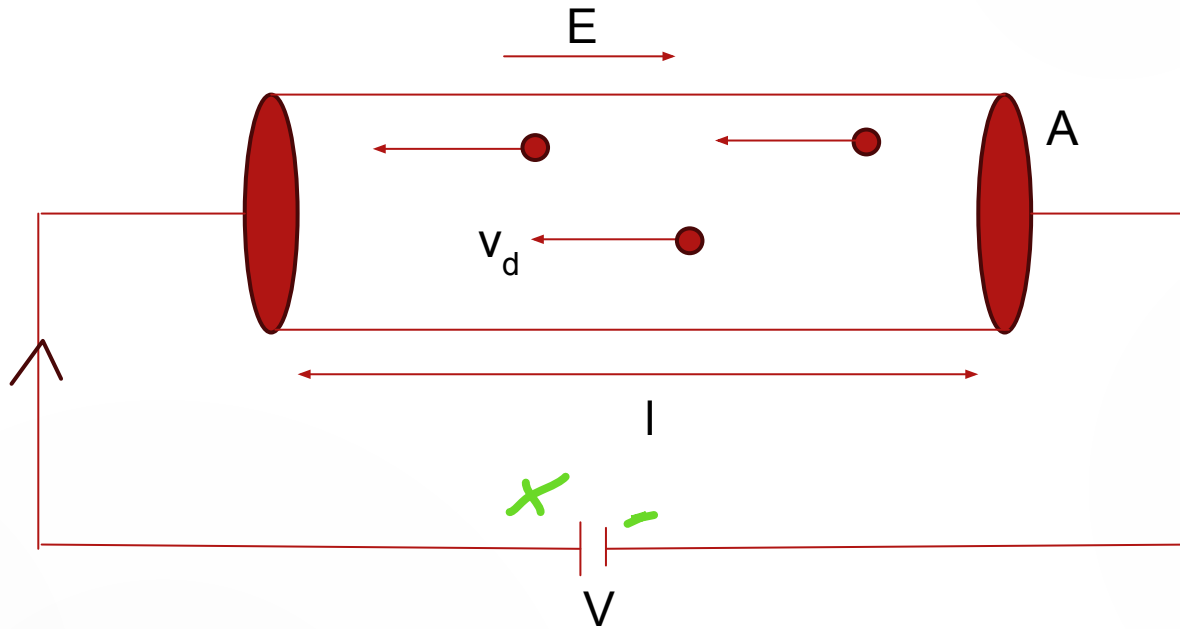




Ohm's Law

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Drift velocity of electrons



Drift velocity is the average velocity of the free electrons in the conductor with which they get drifted towards the positive end of the conductor under the influence of an external electric field.



One of the most basic and important laws of electric circuits is **Ohm's law** named for German physicist Georg Ohm (1789-1854).

Ohm's law states that the voltage across a conductor is directly proportional to the current flowing through it, provided all physical conditions and temperatures remain constant. Mathematically, this current-voltage relationship is written as,

$$V=IR$$

$$V \propto R$$

In the equation, the constant of proportionality, R , is called Resistance and has units of ohms, with the symbol Ω . The same formula can be rewritten in order to calculate the current and resistance respectively as follows:

$$I=V/R$$

$$R=V/I$$

Ohm's law only holds true if the provided temperature and the other physical factors remain constant. In certain components, increasing the current raises the temperature. An example of this is the filament of a light bulb, in which the temperature rises as the current is increased. In this case, Ohm's law cannot be applied. The light bulb filament violates Ohm's Law.

Calculating Electrical Power Using Ohm's Law

The rate at which energy is converted from the electrical energy of the moving charges to some other form of energy like mechanical energy, heat energy, energy stored in magnetic fields or electric fields, is known as **electric power**. The unit of power is the watt. The electrical power can be calculated using Ohm's law and by substituting the values of voltage, current and resistance.

When the values for voltage and current are given,

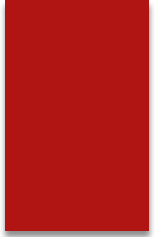
$$P = VI$$

When the values for voltage and resistance are given,

$$P = V^2 \div R$$

When the values for current and resistance are given,

$$P = I^2 R$$



1 Find the resistance of an electrical circuit with a voltage supply of 15 V and a current of 3 mA.

2. Find the current passing through the conductor drawing 20 volts when the power drawn by it is 60 watts.

3. A battery of 6 V is connected to the bulb of resistance 4 Ω . Find the current passing through the bulb and the circuit's power.

4. In a circuit resistors R1 and R2 are in series and have resistances of 5 Ω and 10 Ω , respectively. The voltage across resistor R₁ is equal to 4 V. Find the current passing through resistor R2 and the voltage across the same resistor.

Applications of Ohm's Law in Daily Life

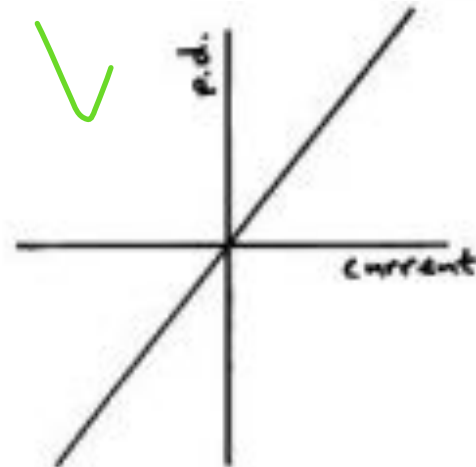
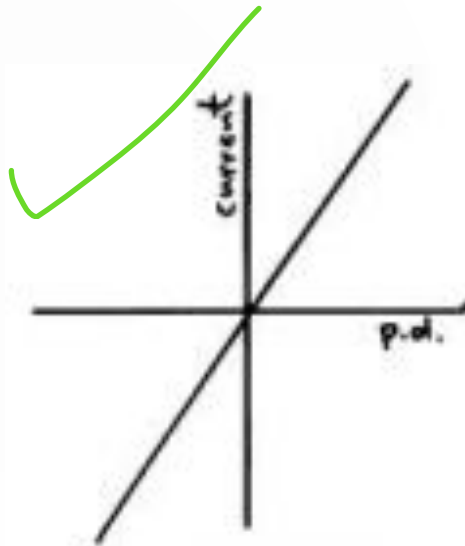
Ohm's Law can determine the voltage applied in a circuit, the value of resistance, and the current flowing through the circuit. With the help of the above three values, we can find the value of other factors like resistivity and many more. Some daily applications of Ohm's law:

- **In fuses:** In order to protect a circuit, fuses and circuit breakers are used. These are connected in series with the electrical appliances. Ohm's law allows us to find the value of the current which could flow through the fuses. If the current value is too large, then it could damage the circuit and even lead to the explosion of the electronic device.
- **To know power consumption:** The electrical heaters have a high-resistance metal coil that allows a certain amount of current to pass across them to provide the heat needed. Using this law, the power to be given to the heaters is determined.
- **To control the speed of fans:** By shifting the regulator to the end from the start, we can regulate the speed of the fans in our houses. By controlling the resistance via the regulator, the current flowing through the fan is managed here. We can measure the resistance, current, and thus power flowing via Ohm's Law for any particular value of the input.
- **For deciding the size of resistors:** Electric appliances like electric kettles and irons have a lot of resistors inside them. In order to provide the necessary amount of heat, the resistors restrict the amount of current that can flow through them. By using Ohm's law, the size of resistors included in them is defined.

Limitation of Ohm's Law

Ohm's law is followed by many substances under specific conditions, however, it is not a fundamental law.

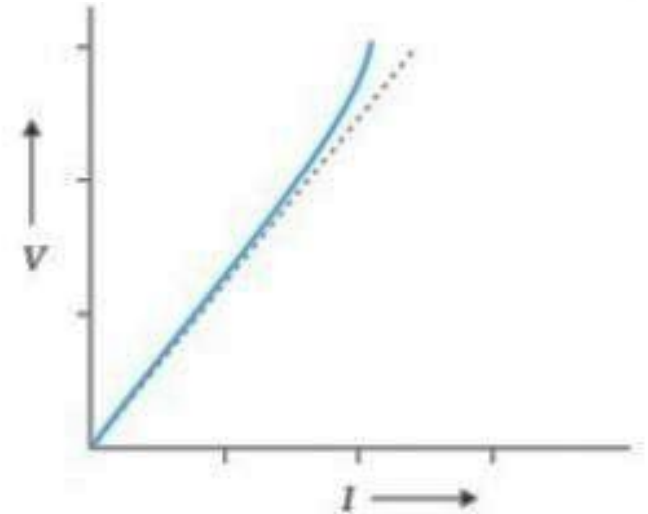
The conductors that follow Ohm's law are called **Ohmic conductors**. For these conductors, **the linear relationship between voltage and current holds true**, i.e. V is proportional to I . The resistance is independent of the current I through the conductor. In these conductors, the current I gets reversed in direction when a potential difference, V , is reversed, but the magnitude of current changes linearly with voltage. Therefore, the V - I graph will be a straight line passing through the origin.



The conductors that don't obey Ohm's law are called **non-ohmic conductors**. The resistance of such conductors is not constant even at a constant temperature. Some examples can be:

1. **Metallic conductors**, which have a straight graph up to a certain current (acts as ohmic) but become non-ohmic at larger currents.

2. **P-n junction diode**, which consists of p-type and n-type semiconductors. A voltage is applied across the junction, resulting in a current I . I is not proportional to V . Very little currents flow for the fairly high negative voltage and a current begins to flow for a much smaller positive bias. Thus, the junction diode allows current to flow only in one direction, i.e, it acts as a rectifier.



Some other examples of non-ohmic conductors are incandescent bulbs, semiconductors- like Germanium and Silicon.

The bulb's tungsten filament heats up when the voltage increases, changing the resistance of the wire. So the bulb filament does not obey ohm's law.